

1. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}: [-\pi, \frac{5\pi}{8}] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) := \left(2 \sin(t - \pi) + \frac{\pi}{2}, \cos(\pi + t), -\sqrt{3} \sin\left(\frac{\pi}{2} - t\right) \right), \quad -\pi \leq t \leq \frac{5\pi}{8}.$$

- (a) Show that $\mathcal{C}$ is a regular curve.
- (b) Show that $\vec{r}'(t) = (-2 \cos t, \sin t, \sqrt{3} \sin t)$.
- (c) Find the curvature κ of the curve $\mathcal{C}$ at the point $(\frac{\pi}{2}, -1, -\sqrt{3})$.
- (d) Write down the cartesian equation of the normal plane to the curve $\mathcal{C}$ at the point in (c).
- (e) Find the length of the arc of the curve $\mathcal{C}$ from the point $(\frac{\pi}{2}, 1, \sqrt{3})$ to the point $(\frac{\pi - 4}{2}, 0, 0)$.

2. Consider the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) := \begin{cases} \frac{\sin^4(x) + \sin^4(y)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0), \end{cases} \quad \forall (x, y) \in \mathbb{R}^2.$$

- (a) Study the continuity of f at $(0, 0)$. (Hint: $|\sin t| \leq |t|$ for every $t \in \mathbb{R}$.)
- (b) Find, if they exist, $f_x(0, 0)$ and $f_y(0, 0)$.
- (c) Say whether the function f is differentiable at $(0, 0)$.

3. Let S be the level surface $S := \{(x, y, z) \in \mathbb{R}^3: -x^2y + z^2 + y - yz = 1\}$. Find the point(s) (x, y, z) on the surface S at which the tangent plane is parallel to the xz -plane.

4. Let $\begin{cases} D_1 := \{(x, y) \in \mathbb{R}^2: x^2 + (y + 1)^2 \leq 1, \quad y \leq -1\} \\ D_2 := \{(x, y) \in \mathbb{R}^2: 1 \leq (x - 2)^2 + (y - 1)^2 \leq 4, \quad y \geq 1\}. \end{cases}$

Show that

$$\iint_{D_1} \frac{y}{x^2 + y^2} dx dy = -1 \quad \text{and} \quad \iint_{D_2} x dx dy = 3\pi.$$

5. Let $E := \{(x, y, z) \in \mathbb{R}^3: x^2 + y^2 + (z + 1)^2 \leq 4 \text{ and } z \leq 1 - x^2 - y^2\}$. Evaluate the triple integral

$$\iiint_E (z + 1) dx dy dz.$$

6. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}: [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) := (2 \cos t + 1, 3 \sin t) \quad \forall t \in [0, 2\pi].$$

Consider the vector field $\vec{F}(x, y) := (x + y, 3y)$ for every $(x, y) \in \mathbb{R}^2$. Evaluate the line integral $\int_{\mathcal{C}} \vec{F} \cdot d\vec{r}$ (i) directly and then, (ii) using the Green theorem.

7. Let $\mathcal{S}$ be the surface described as $\mathcal{S} := \left\{ (x, y, z) \in \mathbb{R}^3: z = \frac{1}{\sqrt{x^2 + y^2}}, \quad 1 \leq z \leq 2 \right\}$.

Show that

$$\iint_{\mathcal{S}} \frac{1}{z^4} dS = \frac{\pi}{3} \left[2\sqrt{2} - \frac{17\sqrt{17}}{64} \right].$$

8. Consider the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = -\sin(x - y) - \cos(2y) \quad \forall (x, y) \in \mathbb{R}^2.$$

- (i) Find **all** the critical points of the function f .
- (ii) Determine the nature (i.e., point of local maximum/minimum or saddle point) of each of the points $(\pi, \frac{\pi}{2})$ and $(\frac{7\pi}{2}, \pi)$.